QIRAN JIA

BIOSTATISTICIAN · COMPUTATIONAL BIOLOGY & MACHINE LEARNING SCIENTIST

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SUMMARY

Biostatistician and bioinformatician developing novel statistical and machine learning methods, software, and analytical pipelines for complex, high-dimensional biomedical data, with expertise in multimodal data integration, statistical inference, and deep learning across bulk and single-cell multi-omics, longitudinal cell imaging, and 3D medical imaging

EDUCATION

University of Southern California — Los Angeles, CA
Ph.D. in Biostatistics — Dec. 2026 (expected)

New York University — New York, NY
M.S. in Biostatistics — May 2022

University of California, Santa Barbara — Santa Barbara, CA
B.A. in Statistics and Chemistry — Dec. 2019

EXPERIENCE

Cellanome — San Diego, CA
COMPUTATIONAL BIOLOGY INTERN — Jun. 2026 - Aug. 2026

- Analyzed multimodal single-cell datasets generated on the R3200 platform with CellCage™ technology across diverse experimental settings using rigorous, streamlined, and reproducible workflows to uncover biological insights
- Developed a deep-learning trajectory modeling framework and supporting internal software for analyzing longitudinal live-cell imaging, capturing cellular morphodynamic patterns and revealing novel biological cell states beyond snapshot morphology or endpoint transcriptomics alone
- Applied novel visualization tools, integrative morphodynamic–transcriptomic analyses, and pathway enrichment to characterize dynamic cell states and their associated transcriptomic profiles

University of Southern California — Los Angeles, CA
GRADUATE RESEARCH ASSISTANT, BIOSTATISTICS — Aug. 2022 - Present

- Engineered LUCID, a statistical quasi-mediation framework and R package for multi-omics clustering and inference with high-dimensional exposures and outcomes, achieving 120× faster computation while extending functionalities including variable selection, uncertainty quantification, missing-data handling, flexible integration strategies, and visualization
- Developed Praxis-BGM, a scalable Bayesian transfer-learning framework for Gaussian mixture models (GMMs) leveraging natural-gradient variational inference and informative biological priors to improve stability, accuracy, and interpretability in high-dimensional, small-sample omics, with applications to TCGA and METABRIC data
- Collaborated regularly with biologists, clinicians, and epidemiologists on method development and applications across environmental health, multimodal single-cell omics, and spatial transcriptomics, providing statistical and computational consulting

New York University — New York, NY
GRADUATE RESEARCH ASSISTANT, BIOSTATISTICS — Feb. 2021 - Sep. 2022

- Designed BiTr-UNet, a CNN–Transformer architecture leveraging self-attention and advanced post-processing techniques for 3D brain tumor segmentation, integrating local and global representations and substantially improving segmentation performance
- Co-developed scalable Hidden Markov Random Field (HMRF)- and deep-learning-based spatial multiple-testing frameworks for inference across hundreds of thousands of FDG-PET voxels, improving power while controlling false discovery rate (FDR)

SAMPLE PROJECTS

Praxis-BGM — Bayesian semi-supervised transfer learning for prior-augmented GMM clustering of bulk and single-cell omics using natural-gradient variational inference [repo]

LUCIDus — R package for integrative clustering of multi-omics data with high-dimensional exposures and outcomes, association inference, missing-data handling, variable selection, and visualization [repo]

BiTr-UNet — CNN–Transformer network for 3D MRI brain tumor segmentation, combining local convolutional features with global transformer representations [repo]

DeepFDR — Deep-learning framework for spatial multiple testing and FDR control in large-scale 3D neuroimaging [repo]

SKILLS

Programming Python, R; JAX, PyTorch, NumPy, Scanpy, Seurat, ggplot2; Linux, Git
Statistical Methods Bayesian inference, variational inference, latent-variable and mixture models, mediation analysis, model-based clustering, longitudinal data analysis, survival analysis, multiple testing under dependence
Machine Learning Representation learning, self-supervised learning, computer vision, transformer-based models, transfer learning
Computational Biology Multi-omics integration, transcriptomics, cell imaging and morphology, statistical genetics
Computing GPU/CUDA, Unix & cluster computing environment, scientific software development

SELECTED PUBLICATIONS

Praxis-BGM: Clustering of Omics Data Using Semi-Supervised Transfer Learning for Gaussian Mixture Models via Natural-Gradient Variational Inference.
Jia Q, Goodrich JA, et al.
Bioinformatics, 2026.

A False Discovery Rate Control Method Using a Fully Connected Hidden Markov Random Field for Neuroimaging Data.
Kim T, **Jia Q**, de Leon MJ, Shu H; Alzheimer's Disease Neuroimaging Initiative.
Medical Image Analysis, 2026.

LUCIDus: An R Package for Implementing Latent Unknown Clustering by Integrating Multi-omics Data With Phenotypic Traits.
Zhao Y, **Jia Q**, Goodrich JA, Conti DV.
The R Journal, 2024.

Integrating Multi-Omics with Environmental Data for Precision Health: A Novel Analytic Framework and Case Study on Prenatal Mercury-Induced Childhood Fatty Liver Disease.
Goodrich JA, Wang H, **Jia Q**, et al.
Environment International, 2024.

An Extension of Latent Unknown Clustering Integrating Multi-omics Data (LUCID) Incorporating Incomplete Omics Data.
Zhao Y, **Jia Q**, Goodrich JA, Darst B, Conti DV.
Bioinformatics Advances, 2024.

DeepFDR: A Deep Learning-Based False Discovery Rate Control Method for Neuroimaging Data.
Kim T, Shu H, **Jia Q**, de Leon MJ, et al.
Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR, 2024.

BiTr-UNet: A CNN-Transformer Combined Network for MRI Brain Tumor Segmentation.
Jia Q, Shu H.
MICCAI BrainLes Workshop, 2021.

PRESENTATIONS

Praxis-BGM: A Prior-Augmented and Regularized Natural-Gradient Variational Inference on Gaussian Mixture Models
Joint Statistical Meetings (JSM), Nashville, TN, Aug. 2025

BiTr-UNet: A CNN-Transformer Combined Network for MRI Brain Tumor Segmentation
MICCAI BrainLes Workshop, Virtual, Sep. 2021